

OMKAR SHETAKE

Software Engineer | Full-Stack & AI-Integrated Backend Systems | Kolhapur, Maharashtra

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building production-grade, AI-integrated full-stack applications using Java, Spring Boot, and Google Gemini AI. Skilled in system design, distributed systems, REST APIs, WebSocket, and cloud-native architecture. Strong foundation in OOP, SOLID principles, and design patterns. Seeking a Software Engineer role to build scalable, intelligent backend systems.

EDUCATION

B.Tech, Computer Science & Engineering — DYP CET, Kolhapur CGPA: 8.5	2023 – 2027
HSC (XII), Science — AVK Junior College, Kasegaon Percentage: 70.5%	2023
SSC (X), Science — AVK Junior College, Kasegaon Percentage: 77.0%	2021

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Backend Frameworks: Spring Boot 3, Spring AI, Spring Security (JWT), Spring WebSocket, Spring Cloud Gateway, Spring Data JPA/Hibernate

Frontend: HTML5, CSS3, Vanilla JS, Leaflet.js, Chart.js, SockJS, STOMP.js

Databases & Cache: PostgreSQL 16 (pgvector), Redis 7

DevOps & Tools: Docker, Docker Compose, Maven, Flyway, Nginx, Git, GitHub, Swagger/OpenAPI, ZXing

Architecture & Design: Event-Driven Architecture (Apache Kafka), System Design, REST APIs, WebSocket/STOMP, API Gateway

Core Concepts: OOP, SOLID Principles, Design Patterns, Data Structures & Algorithms, Unit Testing (JUnit 5 + Mockito)

Cloud & Platforms: AWS Cloud Architecting, AWS Cloud Foundations, Cloudinary, Gmail SMTP

AI / ML & Data: Google Gemini AI (Multimodal), Spring AI, Vector Embeddings (pgvector), NLP, LLMs, Pandas, NumPy

PROJECTS

StayFinder

Java 21 · Spring Boot 3 · Google Gemini AI · Apache Kafka (KRaft) · PostgreSQL · WebSocket · Cloudinary · Leaflet.js · Docker
github.com/OmkarShetake/StayFinderProject

- Built a full-stack AI-powered Airbnb-clone integrating Google Gemini AI (gemini-2.5-flash) for property description generation, smart price recommendations, natural language search, and review summarization.
- Implemented real-time chat and push notifications via WebSocket (STOMP + SockJS); transactional email alerts via Gmail SMTP.
- Architected an event-driven notification system on Apache Kafka in KRaft mode, using a 3-partition topic keyed by userId for per-user ordering, with retry/backoff error handling before dead-letter routing.
- Secured with JWT authentication, BCrypt, role-based access control (GUEST/HOST/ADMIN), XSS sanitization, and password reset.
- Wrote 19 unit tests (JUnit 5 + Mockito); containerized with Docker Compose (Spring Boot + PostgreSQL + Kafka + Nginx).

MedChain

Java 21 · Spring Boot 3 · Spring AI · Google Gemini · PostgreSQL (pgvector) · Redis · Docker · ZXing · Leaflet.js
github.com/OmkarShetake/Medchain

- Built a full-stack AI-powered fake medicine detection platform using Spring Boot 3 and Spring AI, supporting multi-role users (Admin, Manufacturer, Patient, Chemist) with JWT-based RBAC and email verification.
- Integrated Google Gemini 2.5 Flash (multimodal) via Spring AI for medicine image scanning, symptom checking, drug interaction analysis, and AI-powered fake report analysis via scheduled jobs.

- Implemented semantic medicine search using pgvector (vector embeddings) in PostgreSQL 16, enabling intelligent natural language queries across the medicine catalog.
- Built QR code generation and verification system using ZXing; used Redis 7 for caching verification responses with sub-100ms response time; exposed 38 REST endpoints documented via Swagger/OpenAPI.

EXPERIENCE

Participant — Smart India Hackathon 2024 (SIH)

2024

- Collaborated in a team to solve a real-world problem statement, contributing to both frontend (HTML, CSS, JS) and backend (Python) logic under a time-constrained hackathon environment.

Virtual Intern — Cloud & Networking Program (10 Weeks)

2024

- Explored cloud computing concepts, virtual networking, and fundamentals of machine learning using online simulation platforms and lab environments.

CERTIFICATIONS & ACHIEVEMENTS

- AWS Academy Cloud Architecting — Amazon Web Services (AWS), Jan 2025
- AWS Academy Cloud Foundations — Amazon Web Services (AWS), Jan 2025
- Career Essentials in Generative AI — Microsoft & LinkedIn Learning
- Python for Data Science & Data Visualization — CognitiveClass.ai